

CHAPTER 24

OVERREACHING, OVERTRAINING, AND RECOVERY

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After completing this chapter, you will be able to

- define and differentiate overtraining and overreaching;
- understand the difference between the process of overtraining and the result of the overtraining syndrome;
- recognize contributing factors to overtraining and overreaching;
- understand the role of allostatic loading;
- realize that not all decreases in performance are due to overtraining;
- know the physical performances most sensitive to overtraining and overreaching;
- appreciate the physiological, biomechanical, and psychological responses and adaptations to overtraining and overreaching;
- learn about potential strategies for avoiding performance decrements due to overtraining; and
- understand how functional overreaching can be a beneficial part of an appropriately designed training program.

Training for optimum performance is familiar territory for athletes, as it is for most people. Attaining one's performance potential often requires advanced training methods that include phases (i.e., cycles or blocks) of high training volumes and intensities (see [chapter 22](#)). While these training phases can be effective for long-term performance enhancement, they also run the risk of creating a training stimulus that is greater than the recovery capabilities of the individual. When this happens, it is called **overtraining (OT)**, and it contributes to the resulting **overtraining syndrome (OTS)**. The problem of overtraining was recognized as far back as 1866 (101), but the topic of overtraining has received much attention in the scientific literature in recent years, and it is readily apparent that this is a complex and often confusing phenomenon (13, 38, 40, 43, 61, 65, 72, 74, 80, 88, 89, 104, 107, 135, 137, 139). Furthermore, the issue of overtraining is a problem in areas other than sport, such as with tactical (i.e., military) personnel (134). A common finding in many reviews and position stands is that more research must be performed to understand more fully what overtraining and **overreaching (OR)** is, how it affects the individual, and what measures can be taken to avoid this problem. In the present chapter we will examine the known causes and results of the various stress responses as they apply primarily to strength and conditioning programs for high performance and will provide possible strategies to avoid or remedy overtraining syndrome.

Periodization and the General Adaptation Syndrome

When the total training program is correctly designed and implemented, the individual can adapt and improve. This can closely align with Selye's **general adaptation syndrome** (GAS; [figure 24.1](#)) (117). In the GAS paradigm as applied to any biological system, a stress creates an initial alarm phase, followed by a resistance phase, and ultimately adaptation. If the stress is too great, the system is unable to successfully resist the stressor and the biological system dies. In the context of an athlete and a well-designed strength and conditioning program, this sequence initially produces decreases in performance, followed by performance recovery, and eventually performance improvements. This assumes that the training program permits the athlete to resist the stress; otherwise, persistent decreases in performance occur. These performance decrements can be either long term due to overtraining, or short term due to overreaching (see the Overreaching Defined section later in the chapter) (93, 108, 125, 126). Therein lies the problem for the strength and conditioning professional: how to create the proper overload (i.e., training stress) and how to correctly adjust the training load to permit the athlete to recover and attain enhanced performance. When done correctly, this process can be part of a well-designed periodized program (see [chapter 22](#)). It must be remembered that both the GAS and the concept of periodization are paradigms, or models, and do not necessarily have to look exactly like illustrations such as [figure 24.1](#). Indeed, the validity of the GAS as applied to training (14), periodization paradigms (130), and the concept that fatigue can accumulate from resistance exercise programs (63) have been questioned by some academics. However, the vast majority of those who study overtraining and work in the strength and conditioning profession recognize the complexity of this topic (25). It also must

be acknowledged that our understanding of overtraining and overreaching is far from complete (88, 89, 135) and requires much more study. The goal of this chapter is to help the practitioner and athlete avoid unintended long-term decreases in performance, which are exactly the opposite of the intended training goal.

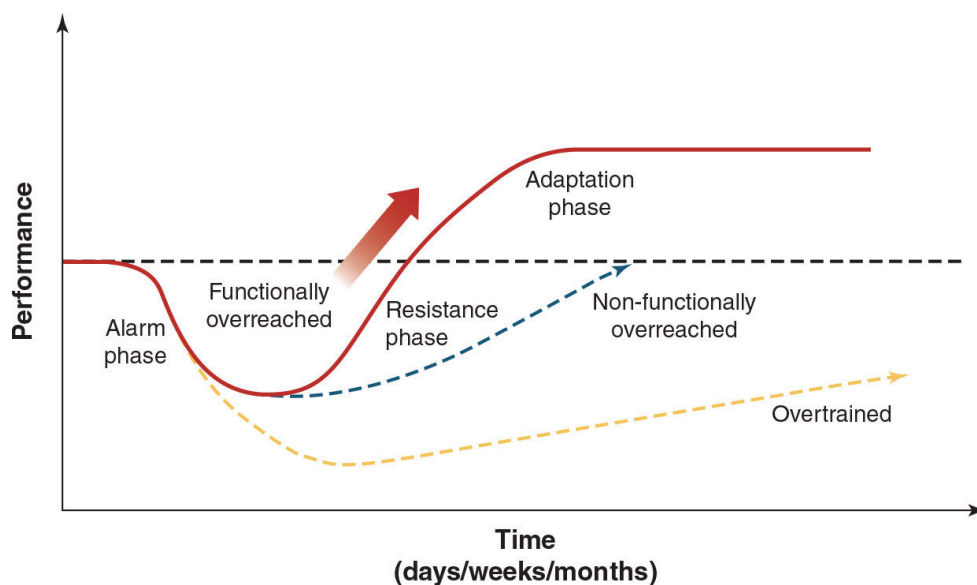


FIGURE 24.1 Relationship between the general adaptation syndrome paradigm and stages of overtraining and overreaching.

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One of the biggest challenges when training an individual or a team is how to get the most from training. If working with athletes training for maximum performance, this means implementing a training program that will optimize performance for their sport. If working in a tactical environment (e.g., military, law enforcement, fire fighter, first responder), the goal again is to optimize performance for the tasks to be performed. Even for those in the general population who are hoping to optimize their health and fitness, it is sometimes desirable to get the maximum effects from the training program to satisfy their needs. In each of these scenarios, the person designing the training program will most likely not be satisfied with mediocre results but is rather seeking to obtain

the best results possible despite potential limitations encountered. These limitations can include a number of factors such as those listed in [figure 24.2](#).

Figure 24.2 Possible limiting factors when designing an optimal strength and conditioning program.

- Time available
- Equipment and facilities available
- Sport-specific training
- Sport governing body regulations
- Prior training history
- Travel and time zones
- Competition density
- Injury and physical limitations
- Menstrual cycle
- Age
- Job
- Family demands and schedule
- Recreational activities

In an effort to maximize the training effect, it may be necessary to include blocks of training that highly tax the individual in an attempt to attain the greatest training response possible. Examples of this can include the high-volume general preparation phases found in many strength and conditioning programs (see [chapter 18](#) for more details). When implemented correctly, the resulting longer-term training effect can be quite substantial. But a problem arises when these phases of training have too high a volume or intensity and are beyond the recovery capacity of the individual or team, which can give way to overtraining. But what exactly is overtraining? What are the lesser forms of overtraining, referred to as *overreaching*? And how do these differ from a stressful but well-designed training block? Although the term *overtraining* is commonly

used in the sport coaching and strength and conditioning professions, we must first define exactly what we are talking about.

Overtraining and Overreaching

A number of definitions of overtraining and overreaching can be found in the literature; however, in many cases the definitions are vague or extremely general. Some may think the concept of overtraining is a relatively new topic. However, as previously mentioned, though not in the context of sport performance, the issue was recognized as early as 1866 concerning general physical effort. It was stated that “the exertion shall not exceed the powers of recruitment; and ... the recruitment shall be facilitated by adequate diet and rest” (101). Additionally, much of the overtraining literature is based on the seminal work of Hans Selye’s GAS (117), which is also discussed in [chapter 22](#). In the 1980s and 1990s, sport scientists and sports medicine clinicians began to recognize the problems of overtraining in sport and began to study and evaluate this challenge as described in several reviews (13, 38, 72, 74, 80, 104, 126, 129, 139). The following things had become clear:

- Not all forms of overtraining are identical (72, 129).
- Not all types of performance decrements are the same (33, 34, 80, 104).
- The contributing factors are variable (13, 38, 72, 80, 104, 126, 129, 139).

[Figure 24.3](#) illustrates the training continuum from no training to optimum training, which can include stressful training to various overtraining scenarios. What should be abundantly clear in this figure is that different sports (i.e., training goals) present different types of overtraining. Likewise, different types of resistance exercise also present different pathways to the problem of overtraining and the resulting OTS. It must be remembered that when proper training programs are followed, the training and performance adaptations can differ considerably (e.g., a distance runner [aerobic endurance sport], wrestler [hybrid sport, both aerobic and anaerobic], or a shot-putter [anaerobic, strength, and power]). Likewise, the factors

contributing to overtraining for these same three athletes, and the performance impairments and physiological responses, are also different (13, 33, 34, 38, 72, 80, 104, 126, 138, 139).

Overtraining Defined

Due to the emergence of numerous attempts to define and understand overtraining (64, 71), a joint task force on the topic was convened by the combined efforts of the European College of Sports Science (ECSS) and the American College of Sports Medicine (ACSM) (88). The result was their position statement titled "Preventions, Diagnosis and Treatment of the Overtraining Syndrome" that presented a consensus definition of overtraining and overreaching, as well as what was scientifically known about these problems at the time. This was followed in 2013 by an updated position stand summarizing what had been learned subsequent to 2006 (89). The consensus of this working group was to adopt the following definition of overtraining:

An accumulation of training and/or non-training stress resulting in *long-term* decrement in performance capacity with or without related physiological and psychological signs and symptoms of maladaptation in which restoration of performance capacity may take *several weeks or months* (88, p. 2).

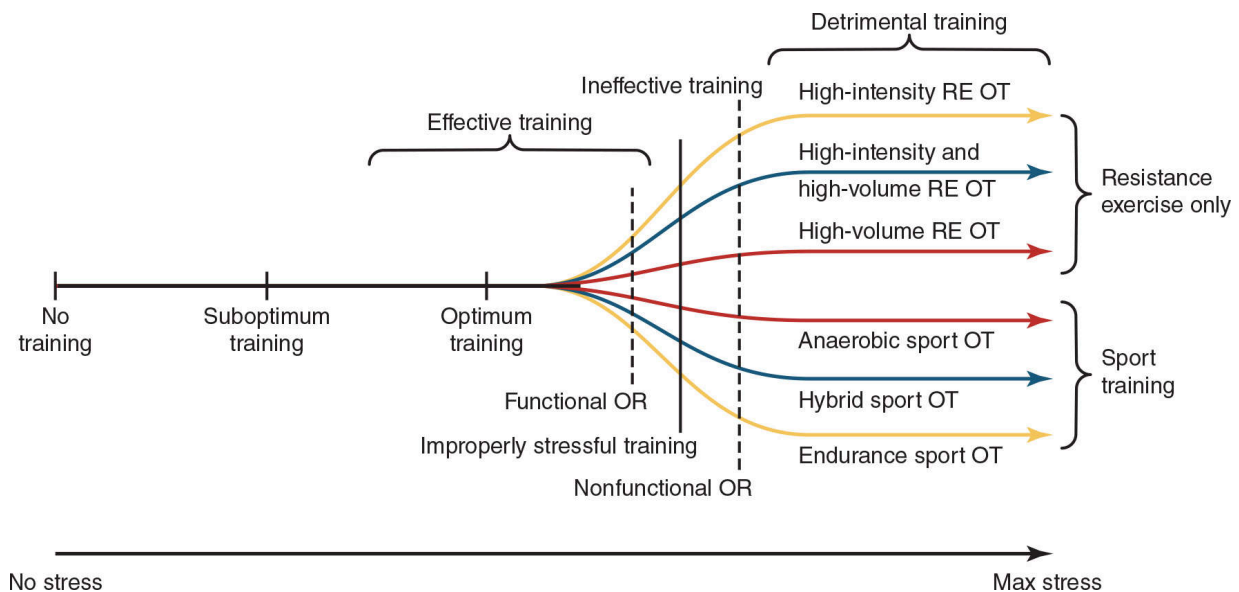


FIGURE 24.3 Training continuum resulting in overtraining for numerous forms of exercise and training. RE = resistance exercise; OT = overtraining; OR = overreaching.

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As can be seen in this definition, this is a serious condition for the athlete from which recovery can take a long time, and it greatly interferes or halts the competition training program. Since this definition was proposed, more recent working groups have added their definitions to the literature, as well as concerns with the state of the overtraining research (65, 135) that have helped illustrate the challenges of understanding this phenomenon.

An important aspect of this definition that was also clarified in these position stands is that overtraining is actually the process (e.g., excessively high volumes and intensities of training) that then can result in an overtraining syndrome (i.e., the collection of symptoms and performance impairments) ([figure 24.4](#)) (88, 89). To be overtrained, one actually has the OTS and the required long-term performance impairments (i.e., weeks to months or longer). Performance impairments are necessary for overtraining to have occurred. However, we will see that not all performance impairments are alike.

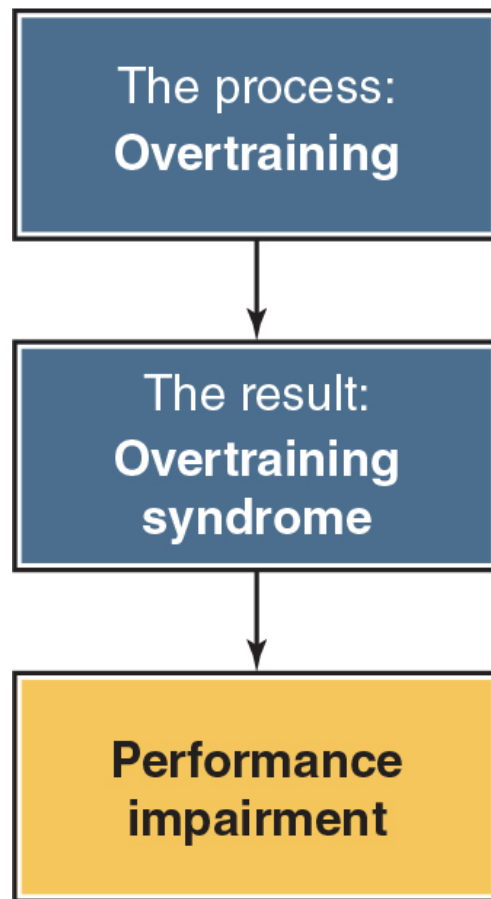


FIGURE 24.4 Overtraining is the process, and the overtraining syndrome is the result. In all instances, performance impairment of some type will be one of the results.

Overtraining is the process that can result in an overtraining syndrome.

Overtraining Syndrome Defined

When the ECSS-ACSM Task Force defined overtraining (88, 89), they specifically stated that the OTS was a combination of “related physiological and psychological signs and symptoms of maladaptation,” some of which may not be part of the training program. As such, the use of the term *syndrome* indicates the complex nature of the condition and the fact that the symptoms are associated with each other.

A major challenge to understanding overtraining and its causes and effects are the numerous different terms found in the literature. This is one reason several groups have developed position stands (88, 89) and consensus statements (65) that include suggested terminology. Likewise, many symptoms of an overtraining syndrome can vary between individuals and between different types of training. However, it is important to note that in all cases, performance is impaired in some manner. [Figure 24.5](#) lists a number of the terms used in the literature for overtraining and its effects.

Figure 24.5 Terminology commonly used for overtraining, the overtraining syndrome, and related topics.

- Overtraining
- Overreaching
- Functional overreaching
- Nonfunctional overreaching
- Overtraining syndrome
- Stagnation
- Overworked
- Overstressed
- Burnout
- Overfatigued
- Excessive fatigue
- Imbalance of work and recovery
- Underrecovered
- Failure to adapt
- Stressful training syndrome
- Unexplained underperformance
- Underperformance
- Staleness
- Maladaptation
- Chronic fatigue

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Research on the OTS sometimes identifies two types of OTS related to the **autonomic nervous system**, specifically the **sympathetic overtraining syndrome** (sometimes referred to as *SOTS*) and the **parasympathetic overtraining syndrome** (sometimes referred to as *POTS*) (126). First presented by Israel in 1976 (61), these syndromes were named based on how the autonomic nervous system responded to the training stresses.

Parasympathetic Overtraining Syndrome

Initially named the **Addisonoid syndrome** since it is often accompanied by decreased concentrations of the stress hormone cortisol, this condition is characterized by the predominance of parasympathetic activity. In general, the parasympathetic overtraining syndrome is most often found in aerobic endurance athletes where very large training volumes are common. Diagnosing parasympathetic overtraining syndrome in aerobic endurance athletes can be tricky since decreased resting heart rate (HR_{rest}) is typically a sign of beneficial cardiovascular adaptations to a training program (e.g., decreased resting HR). However, in this case, the decreased HR in already highly trained individuals is due to a parasympathetic overtraining syndrome and the prerequisite training program. It should be noted that the parasympathetic overtraining syndrome does not cause the performance impairments, but rather is a collection of symptoms typically resulting from the huge training volumes often used by aerobic endurance athletes. Some of the classic studies of the parasympathetic overtraining syndrome can be found in the work of Manfred Lehmann and his research team's study of extremely large training volumes of middle-distance runners (80). Many of the results supported the presence of a parasympathetic overtraining syndrome (80), and the recovery period was as long as a year (personal communication, M. Lehmann). The primary symptom of a parasympathetic overtraining syndrome, besides decreased performance, is a diminished sympathetic response (i.e., **epinephrine**, or **adrenaline**) to

training (78, 80). At times, this has been called **adrenal exhaustion** due to the inability of the adrenal glands to produce adequate concentrations of epinephrine.

Sympathetic Overtraining Syndrome

For a number of years, the existence of a sympathetic overtraining syndrome was hypothetical, until it was shown in a group of resistance exercise overtrained subjects (36). It had been hypothesized that a sympathetic overtraining syndrome would occur before development of a parasympathetic overtraining syndrome, but enhanced sympathetic activity had never been previously shown in response to an OT stimulus. The primary characteristic of sympathetic overtraining syndrome is an increased epinephrine response to exercise and training (36, 37, 38, 40, 41, 139). Eventually, the elevated **catecholamines** result in a **down-regulation** of their primary receptor in skeletal muscle, β -2 receptors (41, 124). This is believed to be an important part of the sympathetic overtraining syndrome and the accompanying impaired contractility of skeletal muscle in this condition.

Overreaching Defined

A lesser form of stressful training than overtraining is called *overreaching (OR)* and is defined as follows:

An accumulation of training and/or non-training stress resulting in *short-term* decrement in performance capacity with or without related physiological and psychological signs and symptoms of maladaptation in which restoration of performance capacity may take from *several days to several weeks* (88, 89).

This definition is almost the same as overtraining; however, the severity of the condition is less, and recovery is faster. It must be appreciated here that there are multiple degrees of performance impairment, including two types of overreaching ([figure 24.6](#)). The most subtle type is **performance stagnation**. While sometimes

stagnated performances have been considered a type of overtraining (13), this is difficult to differentiate from good training programs since all seriously trained individuals experience periodic plateaus in performance. However, if stagnation occurs due to an inappropriate training load (33), adjustments must be made.

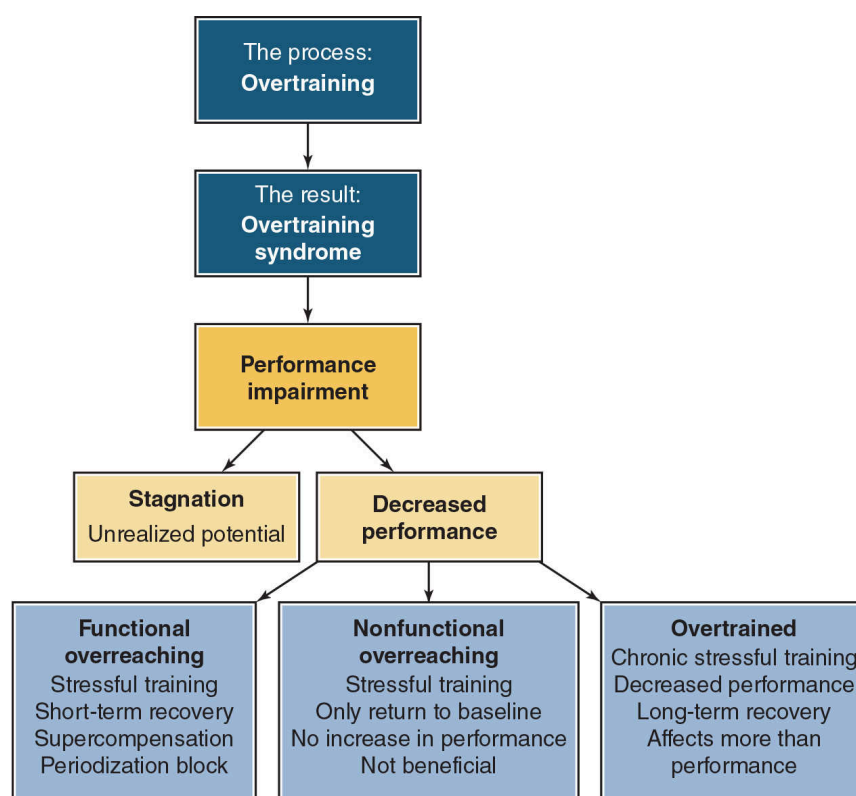


FIGURE 24.6 Different types of performance impairment due to an overtraining syndrome. Being overreached or overtrained results in decreased performance. The primary differences include the recovery times, whether performance is ultimately enhanced, and the magnitude of other symptoms that occur.

Functional Overreaching

Many training programs purposely include micro- or mesocycles of greater training volumes or intensities as part of the long-term plan. Examples of these include preseason training camps for many sports that include two or three training sessions a day or general preparation training blocks in the strength and conditioning program. When properly prescribed, these can be very effective training

methods that prepare the athlete for later performance gains (88, 89, 106, 126). The short-term effect is decrements in performance, but recovery can be attained within days or weeks and can result in a supercompensation effect. As such, and although stressful to the athlete, **functional overreaching (FOR)** can be beneficial provided appropriate recovery is allowed. It should be noted that functional overreaching is a commonly used training strategy by many successful coaches. [Figure 24.1](#) on [page 778](#) illustrates how functional overreaching can fit into the GAS paradigm.

Nonfunctional Overreaching

On the other hand, **nonfunctional overreaching (NFOR)** is somewhat similar to functional overreaching with two very important differences. First, the short-term accumulated fatigue is greater, and, second, the recovery takes longer (88, 89, 126). In the case of NFOR, the individual does not experience any supercompensation and is only able to return to his or her baseline performance levels (see [figure 24.1](#) on [page 778](#)) (88, 89, 93, 108, 125, 126). As the name implies, this type of training does not serve any useful function since no performance benefits result, and the athlete has been unnecessarily stressed.

Overtraining and Overreaching Share Properties With Other Conditions

A number of other problematic conditions share symptoms and characteristics with OT and OTS. For example, similarities between OT and the OTS with clinical depression (4, 5), post-traumatic stress disorder (6), and other relevant psychological conditions (53) have been noted.

Additionally, many characteristics of **relative energy deficiency syndrome (RED-S)**, formerly known as the *female athlete triad*, are similar to symptoms often found in OTS (122). Although the root causes of each of these illnesses or diseases may be different from OT and OTS, they can each certainly contribute to an OTS. Both RED-S and OTS can be interrelated since many causes and symptoms of each condition contribute to the other. Therefore, when the presence of RED-S is suspected, it is important to clearly discern how overtraining may be contributing to this problem. To adequately address the serious condition of RED-S,

trained professionals need to be involved, but the strength and conditioning professional must be able to recognize when help of this type is needed.

Overtraining Continuum

All the different types and severities of overtraining and overreaching can collectively make an overtraining continuum ([figure 24.7](#)). Although some overlap may exist between each of the categories, this continuum illustrates how each successive level becomes more serious for the athlete and requires longer and more intense recovery.

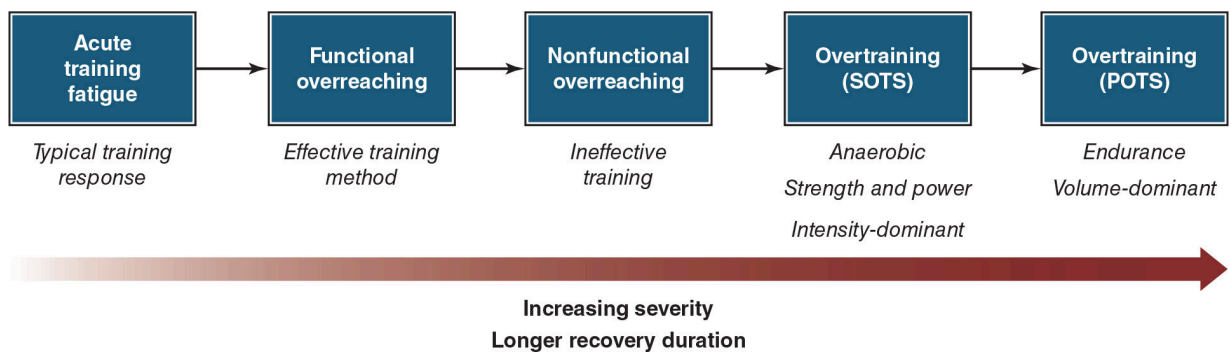


FIGURE 24.7 The overreaching/overtraining continuum. SOTS = sympathetic overtraining syndrome; POTS = parasympathetic overtraining syndrome.

It should be apparent now that not all forms of overtraining are identical (28, 126, 129), and the resulting OTS is not a simple problem with simple solutions. Indeed, it has been repeatedly reported that many factors contribute to this problem, and whether it is due to training for aerobic endurance sports (28, 46) or resistance exercise (37, 43), it is extremely complex (46, 49, 84, 88, 89, 139) and multidisciplinary (75, 107). The net result is that there are many risk factors for overtraining (110, 139) that must be recognized to avoid this problem. As evidenced by some who believe there is no evidence of fatigue accumulation from resistance exercise (63), much more research is needed (48, 135). It is highly likely that understanding this complexity will require novel and emerging research methods and approaches such as advanced statistical

modeling and work in the field of *-omics* (e.g., proteomics and genomics) (4, 30).

Factors Contributing to Overreaching and Overtraining

The first step in understanding overtraining and overreaching is identifying the balance between **stress** and **recovery** (53). Stress from a strength and conditioning program must be countered with adequate recovery to permit long-term progression ([figure 24.8](#)). The inability to adequately recover will undoubtedly result in an overtraining scenario. Likewise, an inadequate training load will likely produce suboptimal performance. The challenge for the strength and conditioning professional is to find the appropriate balance to maximize the long-term performance adaptations.

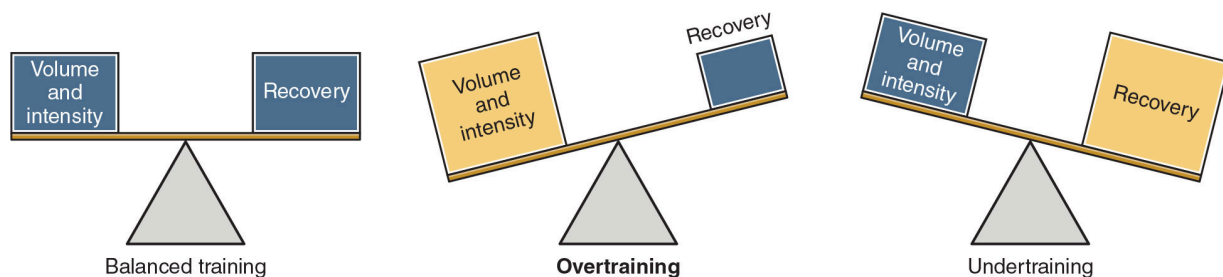


FIGURE 24.8 Although many factors can contribute to poor performance, overtraining, as the name implies, is due to excessive training, inadequate recovery, or both.

Training Factors

Since we are focusing on overtraining and overreaching in the strength and conditioning context, the first and most obvious contributing factors are found in the strength and conditioning program in question. A quick review of the numerous variables in the resistance exercise program illustrates the many ways the program can be adjusted to the needs of the individual being trained (139; see [chapter 18](#)). In general, the volume of training (e.g., total repetitions, volume-load, mechanical work, metabolic work, frequency of training) and the intensity of training (e.g., relative intensity [%RM], absolute intensity [kg], perceptions of effort) are

important variables when determining the training stress ([figure 24.9](#)). Excessive amounts of either training volume or intensity can combine to produce an overtraining stimulus. Monitoring the characteristics of the training program may be one of the most important methods for prescribing load. Additionally, the long-term periodized training program also must be correctly applied. Even though a periodized program can be effective (see [chapter 22](#)), it can also be incorrectly designed, resulting in overtraining. Additionally, an appropriately designed training taper can also enhance performance testing (96). Another training error is a lack of variation in the training load (126), which can be quantified by calculating the **coefficient of variation** (%CV; $SD / \bar{x}$) for the training load (29, 82, 139). Finally, the most important part of an athlete's training stress is the sport-specific training and practices, not to mention the stresses of competition.

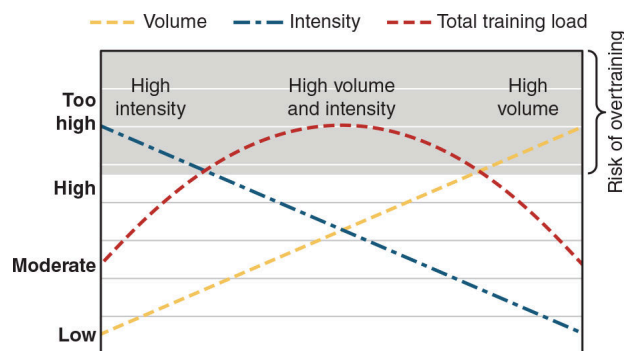


FIGURE 24.9 Excessive levels of training volume and intensity can contribute to overtraining.

Allostatic Load

Often overlooked aspects of an overtraining syndrome are contributing factors outside the training hall or the competition arena. These are many and varied, and in the context of overtraining they have been termed the **allostatic load** (86, 107). Collectively, the cumulative causes of stress (i.e., **antecedents**) produce the various stress responses (i.e., allostatic load) (71, 86). The allostatic load is created by various antecedents (i.e., contributing factors) that

produce adverse responses of the body that are part of the stress responses. To better understand the stress, various algorithms have been created to help make sense of all the antecedents that contribute to this problem (21). In this stress model, if the allostatic load is too great, illness or disease can result. A simple description of the allostatic load is the wear and tear on the body (103). The concept of allostatic load was originally presented in 1993 (86) and is a central concept for the study of how humans respond and adapt to stresses (71). If we now apply the concept of allostatic loads to overtraining syndrome, we see that the totality of stresses to the body include primary factors that are part of the training and recovery process and secondary factors that also contribute to the overtraining syndrome but are not part of training and competition. Collectively, the allostatic load can create a large stress that must be considered by the strength and conditioning professional when planning training programs to avoid overtraining.

[Figure 24.10](#) illustrates how these secondary factors can contribute to an overtraining syndrome. One important note is that these secondary factors that can contribute to the overtraining syndrome can also be caused by an overtraining syndrome. The challenge is to determine if these factors exist outside the training program and add to the overtraining stress or if they are created by overtraining. Many factors have been identified as possibly contributing to overtraining syndrome (107, 110) and are shown in [figure 24.10](#) (1, 8, 26, 29, 31, 34, 36, 37, 39, 40, 41, 45, 52, 54-57, 60, 62, 76, 80, 81, 88, 98, 102, 111, 118-120).

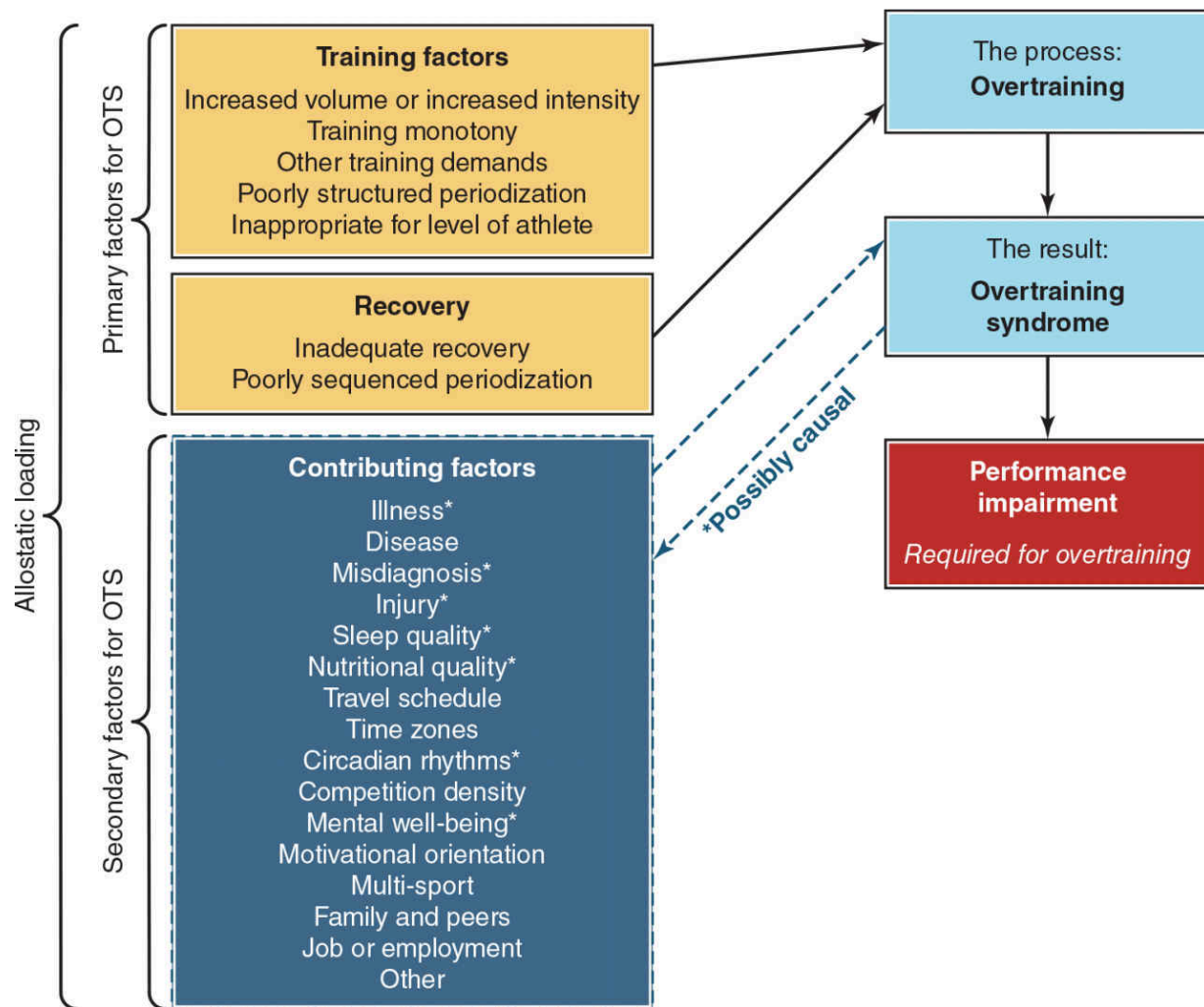


FIGURE 24.10 Allostatic load includes all contributing factors to overtraining and overtraining syndrome (OTS). Solid lines notate required factors or direct causes. Dashed lines show contributing factors, not direct causes.

Underperformance and Its Relation to Overtraining and Overreaching

Despite the name of this chapter, all discussions regarding overtraining and overreaching must start with the concept of **underperformance**. Although all our efforts may go to creating training programs to optimize performance, achieving excellence can be very elusive. Seldom are all components of the training and competition processes in complete synchrony. The job of a strength and conditioning professional is to help navigate the athlete through the many hurdles standing in the way of achieving absolute peak performance. Although few, if any, may ever attain that level of performance perfection, the task of the strength and conditioning professional is to develop, prescribe, and apply the type of training program that gives the athlete the best opportunity to reach that goal. Therefore, it is not only necessary to know how to develop the appropriately challenging training program required to reach this level of performance but also to understand what is happening when performance is less than expected (i.e., underperformance). We will see that even though overtraining and overreaching can be contributors to underperformance, many other factors can contribute as well. Many may be beyond our control, but recognizing possible contributing factors will permit the strength and conditioning professional to determine what appropriate measures can be taken to help the athlete when, despite all the effort, he or she comes up short. In summary, not all cases of underperformance are due to overtraining or overreaching ([figure 24.11](#)). When looking closely at factors that can adversely influence performance, [figure 24.12](#) lists a number of factors that can interfere with optimum performance. As with the stressors that contribute to allostatic load for OTS, many nontraining variables can contribute to underperformance.

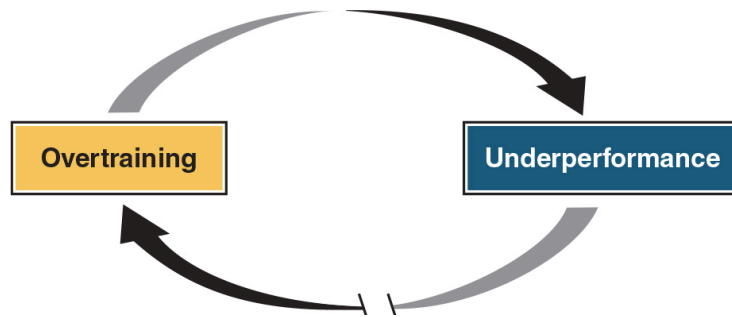


FIGURE 24.11 The results of overtraining are one type of underperformance, but not all types of underperformance are due to overtraining.

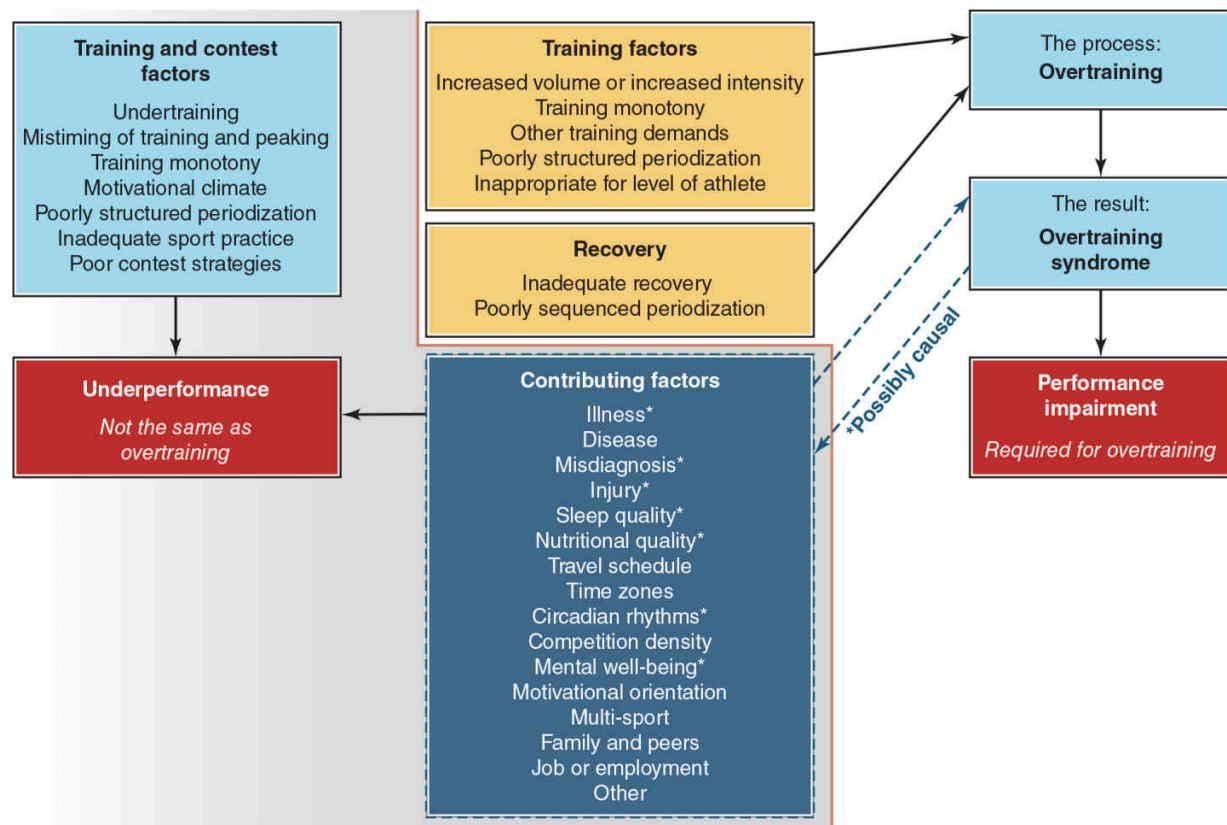


FIGURE 24.12 Underperformance is not necessarily the same thing as overtraining. Solid lines depict direct causes. Dashed lines show contributing factors, not direct causes.

Not all cases of underperformance are due to overtraining or overreaching.

One common training error that can be mistaken for overtraining is due to the timing of the training program with the competition

schedule. [Figure 24.13](#) shows how even a training program that is well designed to produce enhanced performance can be ineffective if the training peak does not coincide with when the athlete needs to perform (96). At first glance, overtraining may be blamed for the underperformance, but the reality is that inappropriate timing, not excessive training, was the cause.

Performance-Related Variables Affected by Overtraining and Overreaching

Up to this point we have discussed what contributes to overtraining and overreaching, but at some point, the strength and conditioning professional needs to know how and when performance is affected ([figure 24.14](#)). To better understand performance decrements due to overtraining, a number of studies have been performed both in the laboratory and in the field (32, 33, 34, 35, 36, 39, 41, 42, 76, 99, 100, 108, 133). One of the problems with this line of research is the difficulty in creating an overtraining syndrome in the laboratory, and when data are taken from actual athletes who have overtrained, the lack of study control makes interpreting the data difficult. Regardless, [table 24.1](#) lists a number of performance measures that have been shown to be adversely affected by overtraining or are believed to be vulnerable to overtraining (18, 33, 35, 36, 38, 39, 41, 42, 43, 50, 52, 78, 105, 106, 124, 139). The wide range of performances that can be affected makes understanding the practical implications of overtraining challenging. In general, when overtraining in the weight room, speed (e.g., sprinting) is most sensitive to overtraining and decreases before most other variables. This is closely followed by decreases in power (force $\times$ velocity), with maximal strength consistently being one of the last performances to decrease. It is essential that the strength and conditioning professionals monitor training performances to determine if any of these performances are changing and to include relevant variables in an athlete monitoring program.

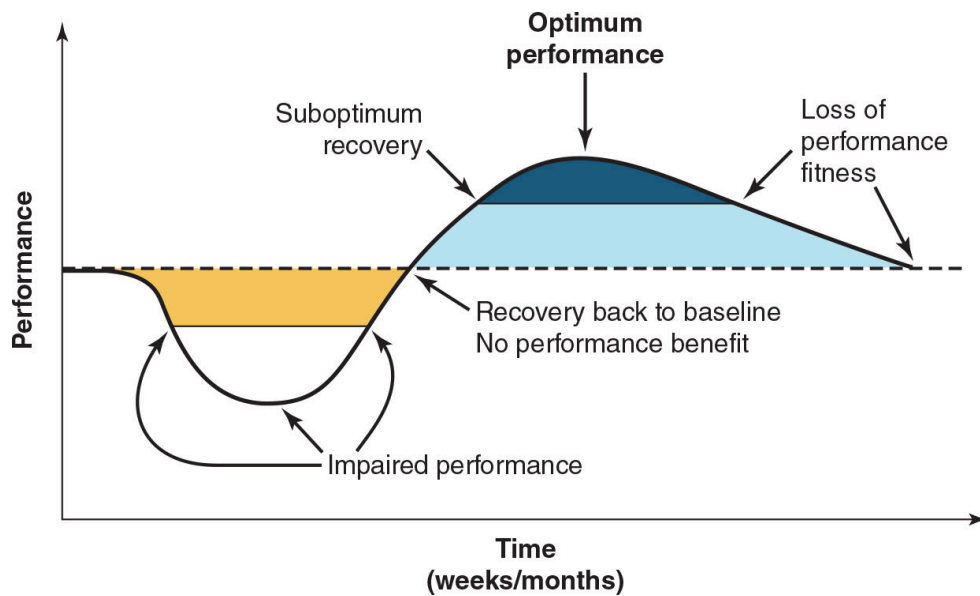


FIGURE 24.13 Training and performance timing. Not all impaired performances are due to overtraining.

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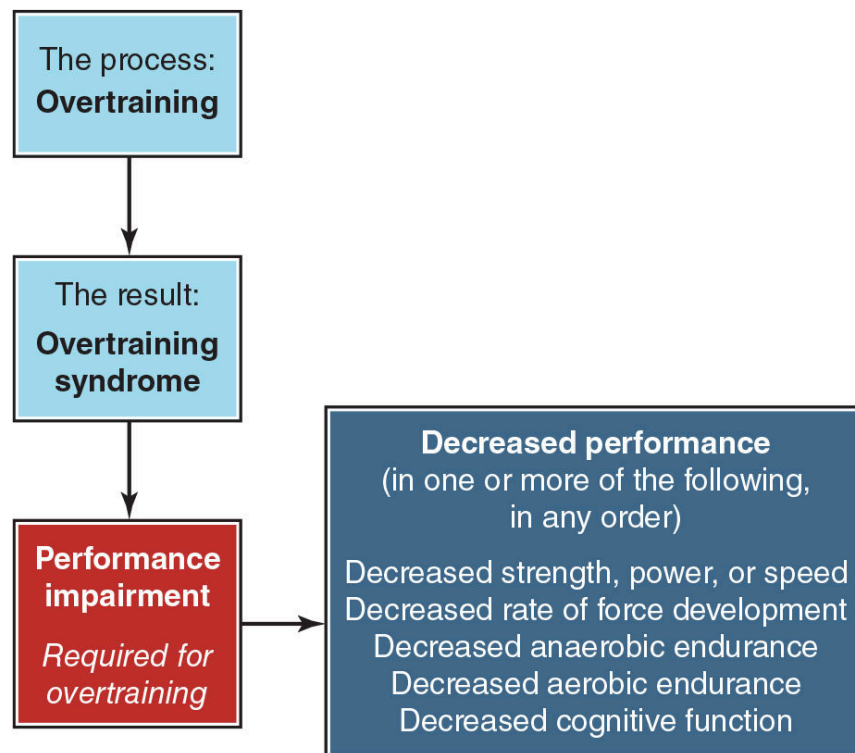


FIGURE 24.14 Influence of overtraining and overtraining syndrome on performance-related variables.

Physiological Mechanisms Associated With Overreaching and Overtraining

As discussed earlier, not all forms of overtraining and the resulting overtraining syndrome are identical. Furthermore, the symptoms and physiological responses can also be quite different depending on the individual and the characteristics of the training program. However, a number of physiological systems have either been shown or theorized to be affected by overtraining. It should be noted that although these mechanisms can be the result of an overtraining stimulus, it is also possible that if these conditions are preexisting, they can also contribute to an overtraining syndrome and be considered part of the allostatic load. Understanding the underlying physiology can, in some cases, provide insight on preventive strategies to help minimize the risk of developing overtraining syndrome and the accompanying impairments in performance. [Table 24.2](#) provides example references for each of the proposed mechanisms, as well as the strength of the data supporting each.

Assessment of Overreaching and Overtraining

Ideally, a carefully designed athlete monitoring program is developed to determine the training and recovery capabilities of each individual. Qualities of the tests included in this program must be specific to the training being performed or the physiological system involved (i.e., valid), reliable (i.e., capable of providing consistent results), sensitive to the training and competition stimulus, and easy to incorporate into the regular athlete monitoring program. Furthermore, those administering the tests must precisely follow correct protocols and must ensure that each person performing the tests provides a maximal effort. If the tests are not administered correctly, the data acquired will be meaningless. Once the tests are performed, an efficient data management system must be used to help best interpret the results rapidly and correctly. Rapid turnaround of test results is important to permit adjustments in the training program when needed.

TABLE 24.1 Performance-Related Variables Affected by Overtraining and Overreaching

Performance variable	Sensitivity	Onset
STRENGTH		
1RM (dynamic constant external resistance)	Low	Last to occur
Isometric force	Poor	Unknown
Isokinetic force	High	Within 1 week
Eccentric strength	Unknown	Unknown
Stimulated force	High	Within 1 week
Single joint	Poor	If isokinetic, 1 week; if DCER, unknown
Multijoint	Moderate	Last to occur

Performance variable	Sensitivity	Onset
RATE OF FORCE DEVELOPMENT		
Single joint	Low	Unknown
Multijoint	Unknown	Unknown
POWER		
Resistance exercise	High	Within 1+ weeks
Vertical jump height	Low	Seldom occurs
Vertical jump kinetics (various forces and reactive strength)	High	Within 1 week
Vertical jump kinematics (positions, strategies)	High	Within 1 week
VELOCITY		
Sprinting	High	Within 1+ weeks
Resistance exercise (DCER)	High	Within 1+ weeks
Change of direction and agility	Moderate	Within 1+ weeks
Efficiency of electrical activity (EMG-to-force ratio)	High	Within days
OTHER VARIABLES		
Self-efficacy (confidence)	High	Within 1 week
Perceptions of recovery	Very high	Within days
Desire to train	High	Within 1 week
Training capacity (toleration of high volumes and frequent sessions)	Moderate	Slower to occur

Performance Testing

[Figure 24.15](#) is a compilation of suggested performance tests for the early detection of an impending overtraining syndrome. Remember, impaired performance is one of the requirements for diagnosing overtraining. While not all inclusive, these tests have either been found to be sensitive to the increased training loads that accompany

overtraining, or it has been theorized that they can provide helpful information (2, 9, 10, 19, 32, 33, 35, 38, 42, 65, 90, 97, 105, 106, 108, 123, 125, 129, 133, 139).

Physiological Testing

Although the strength and conditioning professional does not typically order medical or laboratory tests to monitor the health of his or her athletes and the risk of overtraining, it is helpful to know what types of tests are available and what information they provide. Numerous types of medical and physiological tests have been used to help combat the onset of overtraining syndrome. In general, the meaningfulness of these tests has been equivocal, dependent on the subjects being studied, the training program being performed, and the sport being monitored. A detailed description of all the tests possible is beyond the scope of this chapter, but the reader is welcome to examine some of the following references to get an idea of what is available (9, 10, 19, 34, 36, 37, 39, 40, 45, 50, 54-57, 59, 65, 72, 78-81, 84, 88, 89, 98, 104, 113, 123, 124, 126, 131, 138). In general, numerous potentially helpful variables can be assessed from the blood (e.g., hormones and neurohormones, interleukins, cytokines, markers of muscle disruption or damage, immune factors), cardiovascular measures (e.g., resting heart rate, heart rate variability, oxygen uptake, muscle oxygen saturation), skeletal muscle measures (e.g., heat shock proteins, cell-signaling pathways, receptors), neuromuscular measures (e.g., motor unit recruitment or EMG activity), and orthopedic factors (e.g., joint inflammation or inhibitory input). Data of this type can help the researcher and the practitioner better understand the effects of overtraining and perhaps eventually provide reliable biomarkers of a developing overtraining syndrome.

**TABLE 24.2 Potential Physiological Mechanisms
Contributing to Overtraining Syndrome**

Mechanism	Reference(s)	Evidence
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Mechanism	Reference(s)	Evidence
Sex-related factors	Kelly 2023 (68)	Inconclusive
Circadian rhythm disruption	Nobari et al. 2023 (102)	Likely
Cytokine disruption	Smith 2000 (118) la Torre et al. 2023 (76)	Likely
Glutamine deficiency	la Torre et al. 2023 (76)	Possible
Decreased immune function	Smith 2003 (119) Baker et al. 2022 (8)	Likely
Tissue trauma	Smith 2004 (120)	Possible
Central fatigue theory	Newsholme et al. 1991 (98) Budgett et al. 2008 (15)	Possible
Glycogen deficiency	la Torre et al. 2023 (76)	Strong
Autonomic nervous system disruption	Israel 1976 (61)	Strong
Sympathetic overtraining syndrome (SOTS)	Fry et al. 1994 (36) Fry et al. 2006 (41) Nicoll et al. 2019 (100)	Strong for strength and power training
Parasympathetic overtraining syndrome (POTS)	Lehmann et al. 1991 (78) Lehmann et al. 1999 (80)	Strong for aerobic training
Oxidative stress	Lewis et al. 2018 (81)	Possible
Dietary insufficiencies (e.g., fats, protein, carbohydrate, micronutrients)	la Torre et al. 2023 (76)	Likely
Motor unit recruitment	Fry 1992 (31)	Likely

Mechanism	Reference(s)	Evidence
Endocrine disruption (e.g., testosterone, cortisol, tes/cort, growth hormone, hormone receptors)	Adlercreutz et al. 1986 (1) Fry & Kraemer 1997 (37) Fry et al. 2005 (40) Nicoll et al. 2016 (99)	Strong
Orthopedic inhibition	Fry et al. 2001 (39)	Likely
Muscle cellular signaling	Nicoll et al. 2016 (99) Nicoll et al. 2019 (100)	Strong
Neural modifications	Baker et al. 2022 (8)	Possible
Impaired cognitive function	Symons et al. 2023 (128)	Likely

Figure 24.15 Example performance and training assessments for an athlete monitoring program to minimize risk of developing overtraining syndrome.

Monitored training loads (internal and external)

Resting heart rate

Recovery heart rate

Muscle O₂ saturation

Heart rate variability

Anaerobic thresholds

$\dot{V}O_{2\max}$

RPE/lactate

Muscle O₂ saturation

Ability to repeat a GXT

Ability to complete a training session

Vertical jump performance

Height

Kinetics (forces, power)

Kinematics (positions, strategies)

- Stiffness
- Eccentric use ratio
- Resistance exercise performance
 - Maximal strength
 - Velocity-load profile
 - Power-load profile
 - Bar velocity at same absolute load
- Isometric mid-thigh pulls
- Muscle endurance tests
- Rate of force production
- Sprint performance
 - Acceleration
 - First-step speed
 - Deceleration
 - Max velocity
- Change of direction and agility performance
- Reaction and movement time
- Cognitive processing tests
- Diet records
- Sleep records
- Training questionnaires
- Competition data
- Coaches' subjective assessment

Psychological Measures

Up to this point, we have discussed exclusively quantitative measurements of performance, training, or physiology. However, it has been suggested that questionnaires using subjective measures may be more useful for monitoring athletes and their risk of overtraining (115). To that end, numerous survey instruments have been developed to help monitor athlete perceptions of training status and recovery. [Table 24.3](#) lists a number of these tools that strength and conditioning professionals may find helpful to use. Several considerations when selecting a questionnaire for athlete monitoring include the following:

- How long is the questionnaire, and how many items need to be completed?
- Has the questionnaire been validated?
- Is the wording easy to understand for the intended audience?
- Is it easy to process the data for a rapid turnaround?
- Can the questionnaire be answered without any coercion?
- Is the questionnaire sensitive to an overtraining stimulus?

All of these issues are essential to obtain helpful information. In addition to overtraining-related surveys, several instruments are available to determine the coaching motivational climate (44). It has been clearly shown that the motivational orientation of the athlete as well as the climate created by the coach has a large positive effect on athlete effort, enjoyment, and retention in the sport. These assessments do not need to be administered as often as training-related questionnaires but can be a critical part of the athlete monitoring program that is often overlooked. Needless to say, the coaching climate is highly dependent on coaching strategies. Another related aspect is the mental health and well-being of the coach, which can directly affect athletes' performance satisfaction (7). Finally, it must be realized that the psychological well-being of the athlete may be enhanced by reinforcing mindfulness traits (12, 20), creating both a caring climate and a task-oriented climate (44), and being aware of mood states (95). It stands to reason that mentally healthy athletes may be able to withstand the rigors of stressful training better (64).

TABLE 24.3 Questionnaires for Monitoring Overtraining and Overreaching and the Perceptions of the Psychological Climate

Questionnaire	References
OVERTRAINING RELATED	
Rating of perceived exertion (RPE)	Borg 1982 (11)

Questionnaire	References
CR10 RPE	Borg 1982 (11)
Profile of Mood States (POMS)	McNair et al. 1971 (87) Morgan et al. 1987 (95)
37-item POMS	Coyne et al. 2018 (24)
Daily Analysis of Life Demands for Athletes (DALDA)	Rushall et al. 1990 (112)
Emotional Recovery Questionnaire (EmRecQ)	Lundqvist and Kenttä 2010 (83)
Multi-Component Training Distress Scale (MTDS)	Main and Grove 2009 (84)
Recovery-Stress Questionnaire for Coaches (REST-Q in Sport)	Kellman 2002 (67)
Acute Recovery & Stress Scale (ARSS)	Kellman and Kölling 2019 (67)
Short Recovery & Stress Scale (SRSS)	Kellman and Kölling 2019 (67)
Recovery-Cue Inventory Questionnaire	Kellman et al. 2002 (66)
Total Quality Recovery Questionnaire (TQR)	Kenttä and Hassmén 2002 (70)
Perceived Recovery Status	Laurent et al. 2011 (77)
Rating of Fatigue Scale	Micklewright et al. 2017 (91)
Resistance Exercise Overtraining Questionnaire	Fry et al. (35, 41, 42, 43)
Hooper Index	Hooper et al. 1995 (59)
Training Distress Scale (TDS)	Grove et al. 2014 (51)
Performance Failure Appraisal Inventory (PFAI)	Conroy et al. 2002 (23)

PSYCHOLOGICAL CLIMATE RELATED

Task-Ego Orientation in Sport Questionnaire (TEOSQ)	Fry and Moore 2019 (44)
Perceived Motivational Climate in Sport Questionnaire (PMCSQ)	Fry and Moore 2019 (44)
Caring Climate Scale (CCS)	Fry and Moore 2019 (44)

SLEEP RELATED

Questionnaire	References
Pittsburgh Sleeping Quality Index (PSQI)	Buysse et al. 1989 (17)
Athlete Sleeping Screening Questionnaire (ASSQ)	Samuels et al. 2015 (114)
COACH MENTAL HEALTH RELATED	
Multiday coach survey	Balk et al. 2019 (7)

Recovery Methods and Strategies

Once overtraining or overreaching has occurred and the individual has developed an overtraining syndrome, the most important next step is developing a recovery plan. An important consideration is that overtraining often has a complex and sometimes slow onset and is not rapidly reversed. [Table 24.4](#) lists some of the essential or suggested measures that must be taken when overtraining occurs.

Training Factors for Recovery

Once overtraining occurs and overtraining syndrome has developed, the first thing that must be done is a careful revision of the training program. A rapid decrease in the training volume, intensity, or both is typically warranted. When overreaching has occurred, training cessation has permitted performance to return to original levels in as little as one week (36, 37, 87, 88, 111, 139). When more serious levels of overtraining have occurred, recovery can take as long as four weeks after high-intensity resistance exercise (35) or months to a year after high-volume distance running (78). In each case, training volume or intensity was immediately decreased or removed. It may not always be practical or desirable to cease training, depending on each individual case. Although it has been suggested that one week of resistance training cessation does not benefit strength and hypertrophy gains, the training did not incorporate training similar to overtraining or overreaching (22). Regardless, decreasing or eliminating the training stress is mandatory when overtraining has occurred.

TABLE 24.4 Possible Strategies for Recovering From Overtraining, Overreaching, and the Overtraining Syndrome

Strategy or method	Reference(s)	Effectiveness
TRAINING FACTORS		

Strategy or method	Reference(s)	Effectiveness
Modify training program (e.g., decrease volume, frequency, intensity, training cessation)	Fry et al. 1994 (33) Fry et al. 2006 (41) Nicoll et al. 2019 (100)	Most strong
Training taper (related to above items)	Mujika and Padilla 2003 (96)	Strong

RECOVERY MODALITIES

Mindfulness-based interventions	Cao et al. 2022 (20) Mihajlovic et al. 2023 (92)	Promising
Bed rest	Spiering et al. 2023 (121)	Good for injuries, poor for fitness
Vibration foam rolling	Alonso-Calvete et al. 2022 (3)	Possible
Massage	Li et al. 2024 (82)	Inconclusive
Myofascial release	Martínez-Aranda et al. 2024 (85)	Possible
Quality sleep	Cunha et al. 2023 (26)	Strong
Compression garments	Li et al. 2024 (82)	Possible
Hydration	Mihajlovic et al. 2023 (92)	Possible
Acupuncture	Mihajlovic et al. 2023 (92)	Possible
Red-light therapy	Mihajlovic et al. 2023 (92)	Possible

DIET AND SUPPLEMENTATION

Branched chain amino acids	Salem et al. 2024 (113)	Possible
Protein	Res et al. 2012 (109)	Possible
Carbohydrates	Li et al. 2024 (82)	Promising

Strategy or method	Reference(s)	Effectiveness
Nitrates	Gamonales et al. 2022 (47)	Possible
L-glutamine	Mihajlovic et al. 2023 (92)	Possible
Vitamins	Mihajlovic et al. 2023 (92)	Possible
Oral ATP	Wilson et al. 2013 (136)	Possible

CRYOTHERAPY

Cold-water immersion	Moore et al. 2022 (94) Xiao et al. 2023 (138)	Promising
Other cryotherapy-related methods	Li et al. 2024 (82)	Inconclusive

Recovery Modalities

Many recovery modalities have been successfully used by therapists for medically related conditions. It stands to reason that these methods may also be effective countermeasures for those recovering from overtraining, but there are few data supporting these methods for this purpose. Nevertheless, [table 24.4](#) lists a number of methods that show promise for helping the recovery process.

Integrating Recovery Methods Into the Training Process

Before overtraining occurs, the strength and conditioning professional must be aware of risk factors that can be controlled. The top part of [figure 24.16](#) lists some of these factors, many of which can be readily controlled. The middle part of [figure 24.16](#) lists several items that may be helpful when interpreting physiological data. For more detail on the specifics of these overtraining considerations, the reader is referred to the references listed in [table](#)

[24.4 \(page 792\)](#) and the reviews and chapters cited earlier in this chapter. Finally, the bottom part of [figure 24.16](#) provides action items that can be implemented easily to help the practitioner avoid the onset of overtraining and overtraining syndrome. Before implementing an athlete monitoring program or a system to diagnose an overtraining syndrome, Bernaciková and colleagues (9) suggest classifying any intended tests using three categories of assessments:

- *Category I.* Daily tests, easy to administer, inexpensive, rapid turnaround of results, convenient for the coach and athlete
- *Category II.* Every few days, more complex, longer questionnaires, slightly more inconvenient to the athlete than category I, rapid turnaround of results
- *Category III.* Occasionally administered (e.g., once a month), more involved and intrusive, more expensive, longer turnaround time for the results

Figure 24.16 Overtraining risk factors, considerations, and strategies to minimize risk of developing an overtraining syndrome.

Overtraining Risk Factors

- Lack of close monitoring of the training program (rest days, prior exposure to stressful training)
- Training and activities outside the weight room, which can be a problem
- Use of virtual coaching with no face-to-face contact
- Lack of any objective monitoring
- Training higher level and more fit athletes
- Unmonitored recovery training sessions
- Training during Olympic or world championship training blocks (i.e., final training phase for an ultimate competition)
- Failure to implement an appropriate training taper
- Coaching performed by someone with little coaching experience or training
- Change of training environment (e.g., heat or cold, humidity, altitude)

Overtraining Considerations

- Not all types of overtraining are identical.
- Strength is well-protected.
- Speed and power may be more sensitive to overtraining.
- Psychological variables may be effective markers.
- Hormonal variables indicate stress, not OT.
- Orthopedic problems present different OTS.
- Autonomic nervous system activity responds rapidly to OT.
- Muscle fiber maladaptation can follow.

Overtraining Avoidance Strategies

- Use active and passive recovery methods during training.
- Allow one or more days of routine training abstinence to enhance recovery.
- Rest days are critical.
- Limit excessively long training sessions and too many sessions in a day.
- Exercise caution with excessive training volumes and large increases in training volume (>30% increases).
- Prior exposure to stressful training enhances training stress tolerance.
- Variation in the training stimulus is critical, and use of periodized training is important.
- Avoid excessive training to failure.
- Coordinate resistance exercise with other stressors (e.g., sport training and other conditioning).
- Incorporate recovery strategies (e.g., sleep, diet, mental skills).
- Appropriate diet for training (e.g., adequate Kcal, protein ingestion, glycogen repletion, sufficient vitamins and minerals, adequate fluid intake).
- Dietary supplements may help (e.g., recovery drinks, protein and carbohydrate supplements, creatine, oral ATP).
- Monitor outside stressors.
- Monitor internal and external training loads.
- Incorporate an athlete monitoring program.

It is suggested that category I testing be performed first. If the results suggest possible overtraining concerns, testing for category II is warranted. Likewise, if the category II test suggests an

overtraining problem, a category III test may be performed. In this manner, unnecessary testing can be avoided.

Conclusion

The occurrence of overtraining or nonfunctional overreaching can be devastating to those athletes training for high performance. Not only does it mean they are not able to perform as planned, but it can also mean that training and performance are impaired for an extended period of time. While those who are seriously training may want to push themselves as much as possible, they must be careful not to pass the threshold so that overtraining occurs. Due to the complex nature of this problem, it can result in the maladaptation of many physiological systems ([figure 24.17](#)). Conversely, preexisting problems in any of these physiological systems can also contribute to an overtraining syndrome. Overall, we have discussed in this chapter how everything revolves around the impaired performances caused by improper training ([figure 24.18](#)). However, it is clear that many factors can contribute to this problem. Knowledge of the causes and effects of overtraining and how to avoid it and recover from it are valuable tools for every successful strength and conditioning professional.

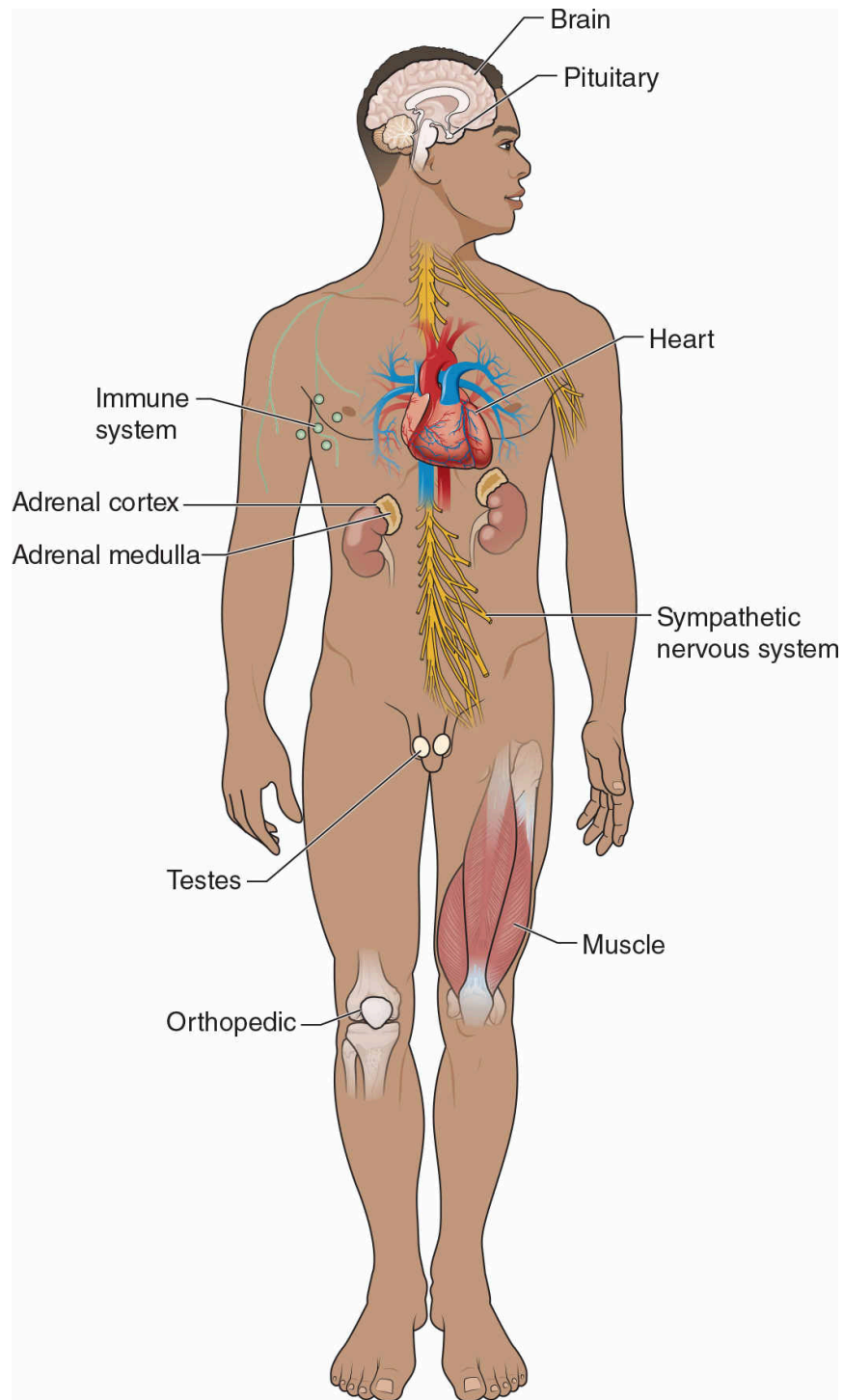


FIGURE 24.17 Overtraining syndrome can affect numerous physiological systems that in turn can contribute to impaired performance.

Adapted by permission from A.C. Fry, "The Role of Training Intensity in Resistance Exercise Overtraining and Overreaching," in *Overtraining in Sports*, edited by R.B. Kreider, A.C. Fry, and M.L. O'Toole (Human Kinetics, 1998), 107-130.

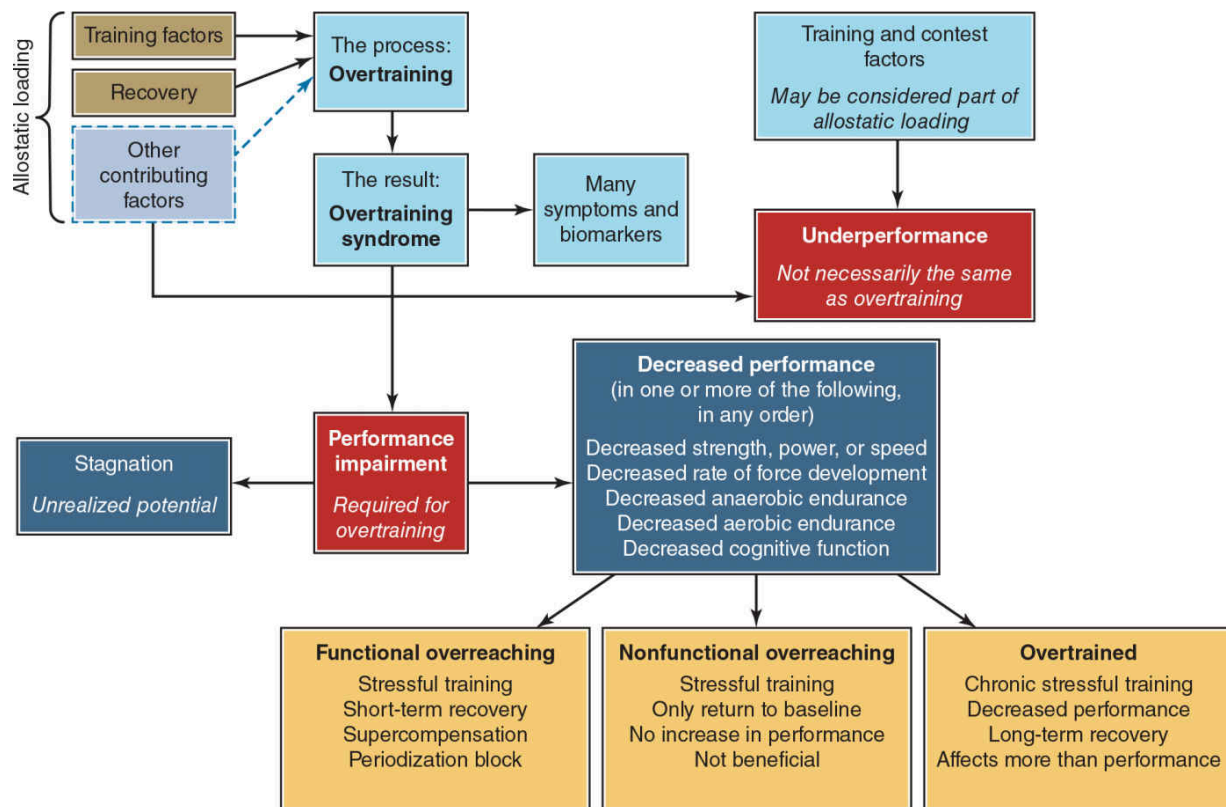


FIGURE 24.18 Summary of factors contributing to overtraining syndrome and different levels of performance impairment. Additionally, many factors other than overtraining can result in underperformance.

KEY TERMS

Addisonoid syndrome
 adrenal exhaustion
 adrenaline
 allostatic load
 antecedents
 autonomic nervous system
 catecholamines
 coefficient of variation
 down-regulation
 epinephrine
 functional overreaching (FOR)
 general adaptation syndrome (GAS)

nonfunctional overreaching (NFOR)
overreaching (OR)
overtraining
overtraining syndrome (OTS)
parasympathetic overtraining syndrome (POTS)
performance stagnation
recovery
relative energy deficiency syndrome (RED-S)
stress
sympathetic overtraining syndrome (SOTS)
underperformance

STUDY QUESTIONS

1. Which of the following is required for an overtraining syndrome to exist?
 - a. impaired performances requiring long-term recovery
 - b. decreased testosterone and increased cortisol concentrations
 - c. lack of success in athletic competitions
 - d. accumulation of nontraining or sport stresses
2. What types of physical performances are first affected by overtraining or overreaching?
 - a. high-velocity activities
 - b. maximum strength tasks
 - c. oxygen uptake capabilities
 - d. repeated interval tasks
3. If factors outside of the training program or sport contests are contributing to overtraining syndrome, what is this collection of factors called?
 - a. allostatic load
 - b. psychosocial dysfunction
 - c. program volume-load
 - d. autonomic nervous system dysfunction

4. When correctly implemented, which of the following can be an effective training tool?
 - a. functional overreaching
 - b. nonfunctional overreaching
 - c. allostatic monitoring phase
 - d. kinetic modulation ratio adjustments
5. Monitoring for overtraining avoidance should include which of the following?
 - a. relevant assessments that are easy to administer and have rapid turnaround of results
 - b. detailed cellular and molecular evaluations to determine the causes of impaired performances
 - c. blood sampling for hormone testing of overtraining-related biomarkers
 - d. determination of key training thresholds that all individuals should avoid